Virtual Arrays, the Difference Co-Array and Vectorization

How physical and virtual sensors work, and how to use them
in near-field cell-free XL-MIMO

Literature review and framework note (simple English)

October 25, 2026

What this note answers. (1) Where the virtual co-array idea came from and why people moved to it; (2) what a "sensor" is, and what *physical* and *virtual* sensors mean; (3) how the difference co-array and vectorization work, step by step, with the mathematics; (4) what earlier papers did, including Pinto and de Lamare's work; (5) how it can help our CAPE-based cell-free system — and, honestly, where it cannot help; (6) four new array-structure ideas for the near field.

1 Where the idea came from, and why

The old limit. With N physical antennas in a uniform linear array (ULA), a receiver can separate at most $N-1$ directions. The number of users or sources you can resolve is tied to the number of antennas, and each antenna costs money, space and one RF chain [1].

First attempt (1968). Minimum-redundancy arrays placed antennas at unequal spacings so that the set of *differences* between antenna positions contained as many distinct values as possible [2]. The idea was already there: what matters is not only where the antennas are, but which *distances* between pairs exist.

The breakthrough: use second-order statistics. If instead of the raw received signal we use its *covariance matrix*, the entries of that matrix depend on *differences* of antenna positions. Those differences form a much larger set than the antennas themselves, called the **difference co-array**. Processing in this domain allows a receiver to identify $\mathcal{O}(N^2)$ sources with only N physical antennas [3, 4].

Closed-form designs. Nested arrays [3] gave the first simple construction whose difference co-array is a *hole-free* (contiguous) virtual ULA, with a closed-form optimum. Co-prime arrays [4] followed, with a co-array that has gaps (holes).

The motivation in one line. Physical antennas are expensive; *differences* between them are free. By working in the covariance (second-order) domain we obtain a much larger "virtual" array, and therefore many more degrees of freedom and finer resolution, without adding hardware [3, 4].

What was missing in a plain nested array. A nested array by itself is still only N physical antennas: if you process the received signal *directly* (first-order), you get only N degrees of freedom and you also suffer side lobes from the sparse spacing. The extra $\mathcal{O}(N^2)$ degrees of freedom appear *only* after you move to the covariance domain and build the co-array [3]. This is exactly why people went from "nested array" to "nested array *plus* difference co-array plus vectorization".

2 What is a "sensor"? Physical vs. virtual

In this literature **sensor = antenna element**. A sensor is one receiving element with its own RF chain. So "N sensors" means "N antennas".

	Physical sensor (antenna)	**Virtual sensor**
What is it?	Real hardware on the panel	A *position* that appears as a difference of two physical positions
Does it exist?	Yes, you can touch it	No. It exists only inside the covariance matrix
Costs an RF chain?	Yes	No
Receives energy?	Yes	**No** — it carries no extra signal power
What does it give?	Signal + array gain	Extra *degrees of freedom* and *resolution*

Very important, and often misunderstood. A virtual sensor does *not* collect any new signal energy. It increases the number of *parameters* (directions, ranges, users) that can be identified and sharpens resolution, but it gives *no* extra beamforming gain and *no* extra received SNR [3, 5]. Any claim of a "N^2-antenna array for free" must be read as "N^2 *identifiable parameters*", not "N^2 antennas".

3 How the difference co-array and vectorization work (step by step)

3.1 Step 1 — the physical received signal

Let the array have N antennas at positions $x_1, \dots, x_N$ (in units of $d = \lambda/2$), and let K sources (users) arrive from angles $\theta_1, \dots, \theta_K$. The received snapshot at time t is

$$\mathbf{y}(t) = \mathbf{A}\,\mathbf{s}(t) + \mathbf{n}(t), \qquad \mathbf{y}(t) \in \mathbb{C}^{N\times 1}, \tag{1}$$

Here $\mathbf{y}(t)$ is the vector of what the N antennas measure at time t (one complex number per antenna), so its size is $N \times 1$. $\mathbf{A} = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_K)]$ is the steering matrix: each column $\mathbf{a}(\theta_k)$ says how source k appears across the antennas. $\mathbf{s}(t)$ holds the K source symbols at time t, and $\mathbf{n}(t)$ is the receiver noise, assumed white with power σ^2. The steering vector entries are

$$[\mathbf{a}(\theta)]_n = e^{\jmath \frac{2\pi}{\lambda} x_n \sin\theta}, \qquad n = 1, \dots, N. \tag{2}$$

Here $[\mathbf{a}(\theta)]_n$ is the phase that a plane wave from angle θ produces at antenna n; x_n is that antenna's position and λ the wavelength. Note this is the far-field *(flat wavefront) model — we return to this point in Section 6.*

3.2 Step 2 — the covariance matrix exposes the differences

Assume the sources are uncorrelated with powers $p_1, \dots, p_K$. The covariance is

$$\mathbf{R} = \mathbb{E}\big[\mathbf{y}(t)\mathbf{y}(t)^{\mathsf{H}}\big] = \mathbf{A}\,\mathrm{diag}(\mathbf{p})\,\mathbf{A}^{\mathsf{H}} + \sigma^2\mathbf{I}_N, \tag{3}$$

Here $\mathbf{R}$ is the $N \times N$ covariance matrix (how each antenna's signal correlates with every other antenna's). $\mathbf{p} = [p_1, \dots, p_K]^{\mathsf{T}}$ collects the source powers, $\mathrm{diag}(\mathbf{p})$ *puts them on a diagonal, σ^2 is the*

noise power, and $\mathbf{I}_N$ *is the identity matrix.* Looking at one entry of $\mathbf{R}$ makes the key point visible:

$$[\mathbf{R}]_{m,n} = \sum_{k=1}^{K} p_k \, e^{\jmath \frac{2\pi}{\lambda}(x_m - x_n)\sin\theta_k} + \sigma^2 \delta_{m,n}. \tag{4}$$

Here $[\mathbf{R}]_{m,n}$ *is the correlation between antenna* m *and antenna* n*, and* $\delta_{m,n} = 1$ *if* $m = n$ *and* 0 *otherwise. The phase depends only on the* ***difference*** $x_m - x_n$*.* This is the whole trick: the covariance behaves as if there were an antenna at every difference position. The set of these positions is the **difference co-array**

$$\mathbb{D} = \{\, x_m - x_n \;:\; 1 \le m, n \le N \,\}. \tag{5}$$

Here $\mathbb{D}$ *is the list of all differences between pairs of physical antenna positions. Each element of* $\mathbb{D}$ *is one* virtual sensor.

3.3 Step 3 — vectorization turns the co-array into a "received vector"

Stack the columns of $\mathbf{R}$ into one long vector. Using the identity $\mathrm{vec}(\mathbf{A}\,\mathrm{diag}(\mathbf{p})\,\mathbf{A}^{\mathsf{H}}) = (\mathbf{A}^* \odot \mathbf{A})\mathbf{p}$,

$$\boxed{\mathbf{z} = \mathrm{vec}(\mathbf{R}) = (\mathbf{A}^* \odot \mathbf{A})\mathbf{p} + \sigma^2 \mathrm{vec}(\mathbf{I}_N), \qquad \mathbf{z} \in \mathbb{C}^{N^2 \times 1}.} \tag{6}$$

Here $\mathbf{z}$ *is the vectorized covariance: a single long column of length* N^2*.* $\mathrm{vec}(\cdot)$ *stacks matrix columns on top of each other,* $(\cdot)^*$ *is the complex conjugate, and* $\odot$ *is the Khatri–Rao (column-wise Kronecker) product.* $\mathbf{p}$ *is again the vector of source powers.* The crucial structure is that column k of $\mathbf{A}^* \odot \mathbf{A}$ is $\mathbf{a}^*(\theta_k) \otimes \mathbf{a}(\theta_k)$, whose entries are $e^{\jmath \frac{2\pi}{\lambda}(x_m - x_n)\sin\theta_k}$ — exactly the steering vector of a *virtual array* sitting at the difference positions (5) [3]. So (6) reads like "received signal = virtual steering matrix × source powers + a deterministic noise term".

Size comparison (the professor's point). Regular or nested array: $\mathbf{y} = \mathbf{A}\mathbf{s} + \mathbf{n}$ with $\mathbf{y}$ of size $N \times 1$ — N physical sensors.
Difference co-array + vectorization: $\mathbf{z} = (\mathbf{A}^* \odot \mathbf{A})\mathbf{p} + \sigma^2 \mathrm{vec}(\mathbf{I})$ with $\mathbf{z}$ of size $N^2 \times 1$ — physical *and* virtual sensors together.

3.4 Step 4 — remove duplicates and restore rank

Many differences in (5) repeat, so $\mathbf{z}$ is reduced to the distinct lags, giving a virtual array $\tilde{\mathbf{z}}$ with $|\mathbb{D}|$ elements. Two difficulties then appear:

1. **Holes.** For a co-prime array $\mathbb{D}$ has gaps, so the virtual array is not a contiguous ULA; the gaps must be filled by interpolation (for example nuclear-norm minimisation) or avoided by design [7]. *Nested arrays avoid this: their difference co-array is hole-free* [3].
2. **Rank-one problem.** In (6) the "signal" $\mathbf{p}$ is deterministic, so $\tilde{\mathbf{z}}$ behaves like a *single snapshot* of fully coherent sources and subspace methods such as MUSIC fail [5].

The classical fix is **spatial smoothing**: split the virtual ULA into overlapping subarrays and average their covariances,

$$\mathbf{R}_{\mathrm{ss}} = \frac{1}{L_v} \sum_{i=1}^{L_v} \tilde{\mathbf{z}}_i \tilde{\mathbf{z}}_i^{\mathsf{H}}, \tag{7}$$

Here $\tilde{\mathbf{z}}_i$ *is the portion of the virtual array vector taken by the* i*-th subarray (a sliding window),* L_v *is the number of such subarrays, and* $\mathbf{R}_{\mathrm{ss}}$ *is the smoothed covariance, which is full rank and can be given to MUSIC.* Liu and Vaidyanathan later showed the smoothing step is not strictly necessary: the same noise subspace can be obtained directly from an easier-to-compute matrix [5].

3.5 The nested array geometry (the one used in this literature)

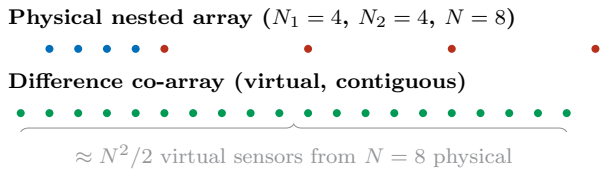
A two-level nested array is the union of two ULAs: an inner one with N_1 antennas at spacing d, and an outer one with N_2 antennas at spacing $(N_1+1)d$ [3]:

$$\mathbb{S} = \underbrace{\{\, m\,d \;:\; m = 1, \ldots, N_1\}}_{\text{inner, dense}} \cup \underbrace{\{\, n(N_1+1)d \;:\; n = 1, \ldots, N_2\}}_{\text{outer, sparse}}. \tag{8}$$

Here $\mathbb{S}$ is the set of physical antenna positions, N_1 and N_2 are the numbers of antennas in the inner and outer parts, and $d = \lambda/2$. The total is $N = N_1 + N_2$ antennas. Its difference co-array is a *contiguous* virtual ULA running from $-(N_2(N_1+1)-1)$ to $+(N_2(N_1+1)-1)$, i.e.

$$|\mathbb{D}| = 2N_2(N_1+1) - 1 \;\approx\; \tfrac{N^2}{2} \quad \text{virtual sensors,} \tag{9}$$

Here $|\mathbb{D}|$ is how many virtual sensors the co-array contains. With $N_1 = N_2 = N/2$ this is about $N^2/2$: from N real antennas we obtain roughly $N^2/2$ virtual ones.



4 What earlier papers did

Work	Contribution
Moffet [2]	Minimum-redundancy arrays: maximise the distinct differences; no closed form.
Pal & Vaidyanathan [3]	**Nested arrays**: closed-form two-level design whose difference co-array is a hole-free virtual ULA; $\mathcal{O}(N^2)$ degrees of freedom from N sensors using second-order statistics.
Vaidyanathan & Pal [4]	**Co-prime arrays**: two co-prime ULAs; co-array has holes but the design is flexible.
Liu & Vaidyanathan [5]	Analysed the spatial-smoothing step in co-array MUSIC and showed an equivalent, cheaper construction (no smoothing needed).
Liu *et al.* [7]	Co-array *interpolation* (nuclear-norm minimisation) to fill the holes of co-prime co-arrays.
Liu & Vaidyanathan [6]	**Super nested arrays**: keep the hole-free co-array but reduce *mutual coupling* between close antennas.
Sum/difference designs [8, 9]	Use both *sum* and *difference* lags (e.g. for non-circular signals) for still more degrees of freedom.
Near-field variants [10, 11, 12]	**Symmetric** nested arrays (SNA, compressed SNA, symmetric double-nested) for near-field and mixed near/far-field localisation — see Section 6.
Pinto & de Lamare [13]	**Co-array for communications** (not only DOA): uplink multiuser MIMO with nested/co-prime arrays; MMSE and SIC receivers on the difference co-array; sum-rate analysis.

4.1 Pinto and de Lamare's work in detail

The paper is *"Study of Multiuser Multiple-Antenna Wireless Communications Systems Based on Super-Resolution Arrays"* by S. F. B. Pinto and R. C. de Lamare [13] (arXiv:2312.06033, December 2023; a related version appeared at EUSIPCO 2023, Helsinki, in the *MIMO and Array Processing* session). From its abstract and section outline:

- **Setting:** the *uplink* of a multiuser system where users transmit to a multiple-antenna base station equipped with *super-resolution* (sparse) arrays.
- **Arrays:** *co-prime* and *two-level nested* arrays, handled by one common procedure.
- **Receivers:** linear MMSE receive filters, plus linear and *successive interference cancellation* (SIC) receivers, designed to process the signals *of the difference co-array*.
- **Analysis:** closed-form expressions for the achievable sum-rates.
- **Claims:** the proposed systems outperform standard-array systems that use *more* antenna elements, with gains in achievable rate and bit-error rate and savings in antennas and energy.

Honest limitation of this review. The network in this environment blocks direct downloads from arXiv and the publisher sites, so I could read only the abstract, section outline and indexed summaries of [13] — **not** the full derivations or the numerical results. The description above is therefore faithful to the abstract but the exact receiver derivation and simulation parameters must be checked against the paper itself before you reproduce or cite them.

Which technique to adopt for comparison. For our purpose the right baseline is exactly [13]: a **two-level nested array** (8) **with the difference co-array and vectorization** (6), because (i) it is the version designed for *communications* (MMSE/SIC receivers, sum-rate, BER) rather than pure DOA, (ii) it comes from the same group as our work, and (iii) the nested geometry gives a hole-free co-array, so no interpolation step is needed.

5 How this can help our work — and where it cannot

Our pipeline is: pilots → per-AP angle/range estimate → CAPE cross-AP position fusion → channel reconstruction → cross-AP list SIC detection. The co-array belongs in the *first* step.

Where it helps	**Why**
Sharper per-AP parameter estimates	The virtual aperture is much larger than the physical one, and resolution improves with aperture, so the angle/range candidates fed to CAPE are far more accurate [3].
More identifiable users	Up to $\mathcal{O}(N^2)$ sources instead of N — useful when the number of users per AP is large [3, 4].
Fewer RF chains	The same resolution with fewer physical antennas, i.e. lower hardware cost [13].
Side-lobe robustness	The co-array processing works on lags, which is a different and often more robust way of handling the ambiguity that worries us with sparse arrays [5].

Four honest limitations you must plan for.

1. **No extra energy.** Virtual sensors add degrees of freedom and resolution, *not* received power. They cannot improve the beamforming gain of the data detector by themselves [3].
2. **Needs many snapshots.** The covariance $\mathbf{R}$ in (3) must be *estimated* from data. Our pilot phase gives only $\tau = K$ snapshots, which is very few, so $\hat{\mathbf{R}}$ is noisy and the co-array gain shrinks [5].
3. **Degrees of freedom are not our bottleneck.** In our setting $K = 8$ users and $N = 63$ antennas per AP, so $K \ll N$: we do *not* need $\mathcal{O}(N^2)$ degrees of freedom. The benefit for us would be *resolution*, not source count.
4. **The near field breaks the standard co-array** — see Section 6. This is the most serious issue for our system.

6 The near-field catch (critical)

Equation (2) assumed a *flat* wavefront. In the near field the wavefront is curved, so the phase contains an extra range-dependent (quadratic) term and the amplitude also varies across the array. Consequently the neat cancellation that produced a clean virtual ULA in (4) no longer holds: the ULA behaviour of the difference co-array *breaks down* for near-field sources because of curved wavefronts and range-dependent attenuation [10, 11].

The known remedy: symmetry. If the array is made *symmetric* about its centre, the quadratic range term cancels in particular lag combinations, which restores a usable co-array and even allows the range to be recovered. This is why the near-field literature uses *symmetric* nested arrays (SNA), compressed symmetric nested arrays (CSNA), symmetric double-nested and symmetric extended nested arrays [10, 11, 12]. A typical procedure is: estimate the angles by spatial-smoothing MUSIC on the difference co-array, then obtain each range by a one-dimensional peak search using the estimated angle [11].

So, for our hybrid near/far-field system: a plain two-level nested array with the standard difference co-array is *not* directly valid on the near-field links. Either restrict the co-array processing to the far-field links, or adopt a *symmetric* structure. Our current ECA is already symmetric about its centre, which is a useful starting point.

7 Four new structure ideas for the near field

These are proposals, not results. Idea 1 is closest to existing work; Ideas 2–4 are, to my knowledge, not published for near-field cell-free systems — verify before claiming novelty.

Idea 1 — Symmetric nested co-array per AP (safest). Replace the ECA with a *symmetric* two-level nested array so the difference co-array stays valid in the near field [10, 11]. *Benefit:* hole-free co-array *and* near-field validity. *Risk:* symmetric nested arrays already exist, so the novelty is only in the cell-free application.

Idea 2 — Cross-AP (distributed) co-array. Form the difference co-array *across APs*, using the union of all AP antenna positions, instead of inside one AP. Since CAPE already fuses several APs, the lag set becomes enormous and its "virtual aperture" spans the whole deployment. *Benefit:* resolution limited by the *network* size, not the panel size; it matches CAPE naturally. *Risk:* APs are not phase-synchronous, so only non-coherent (power-domain) lags may be usable — this must be checked first.

Idea 3 — Joint sum-and-difference co-array for the near field. Near-field phase contains both a linear (angle) and a quadratic (range) term, which naturally produce *sum* lags $x_m + x_n$ as well as difference lags. Designing the geometry so that *both* co-arrays are hole-free would give joint angle–range degrees of freedom in one step. Sum-difference designs exist for far-field and non-circular signals [8, 9]; extending them to near-field cell-free appears open.

Idea 4 — Hybrid-field reconfigurable co-array. Our network is hybrid: some links are near-field, some far-field. Use the *full* difference co-array on far-field links (where it is exactly valid) and a *symmetric sub-array* with range-aware processing on near-field links, switching per link using the Rayleigh-distance test we already compute. *Benefit:* each link uses the processing that is actually valid for it.

For the code comparison you asked for, we use *no* new idea: we implement the published technique of [13] — a two-level nested array (8) with the difference co-array and vectorization (6) — and compare it against our ECA-based CAPE system on BER, sum-rate, estimation error and localisation.

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